

Man-in-the-Middle attack. **Analyze DVWA HTTP Login Using Wireshark**

Question:

Demonstrate, using Wireshark, that HTTP transmits login credentials in plaintext by capturing and analyzing a DVWA login request.

Step 1: Start DVWA

Open DVWA on your own machine or lab VM.

Example: `http://<metasploitable-ip>/dvwa`

```
Login with msfadmin/msfadmin to get started
```

```
metasploitable login: msfadmin
Password:
Last login: Sun Jul  5 03:03:50 EDT 2026 on tty1
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686
```

```
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
```

```
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
```

```
To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
```

```
msfadmin@metasploitable:~$ ip addr | grep 192\.\168\.\74
    inet 192.168.74.130/24 brd 192.168.74.255 scope global eth0
msfadmin@metasploitable:~$
```

Damn Vulnerable Web App (DVWA) x +

Not secure 192.168.74.130/dvwa/login.php

DVWA

Username

Password

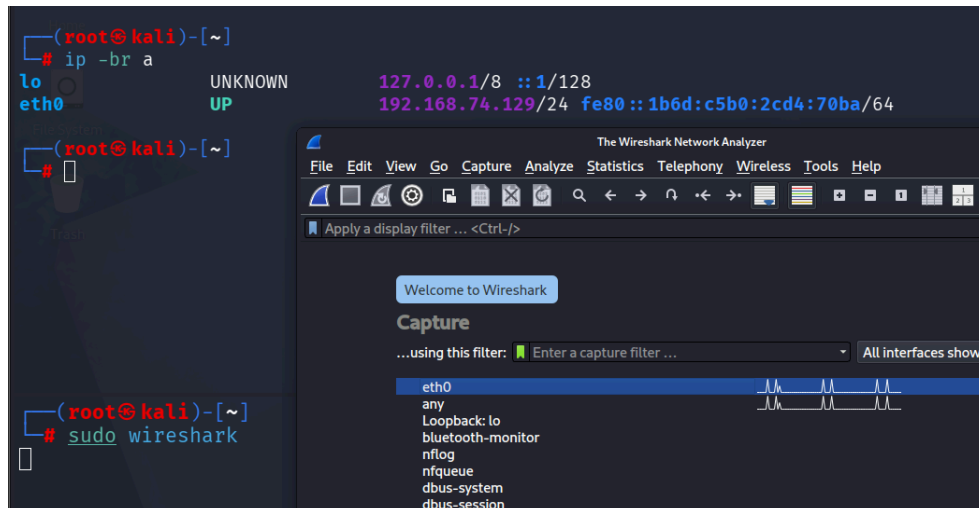
Login

Damn Vulnerable Web Application (DVWA) is a RandomStorm OpenSource project

Hint: default username is 'admin' with password 'password'

Step 2: Launch Wireshark

1. Open **Wireshark**.
2. Select the network interface connected to your lab network (for example, **eth0** or **ens33**).
3. Right Click **Start Capturing Packets**.

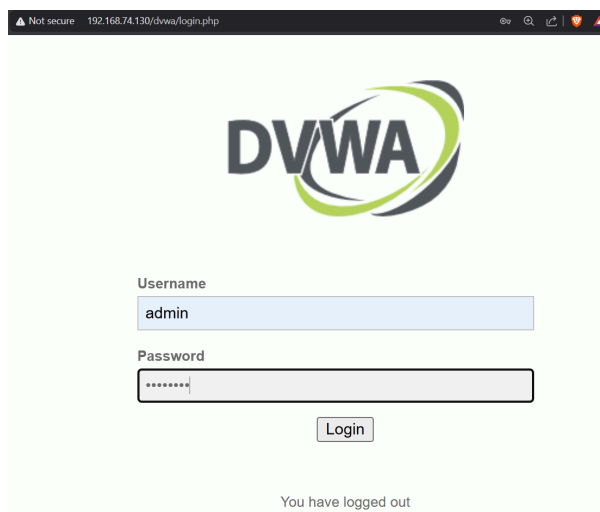


Step 3: Generate HTTP Traffic

From the same machine being captured:

- Open the DVWA login page.
- Enter the default credentials.
- Click **Login**.

This creates an HTTP POST request that Wireshark can capture.



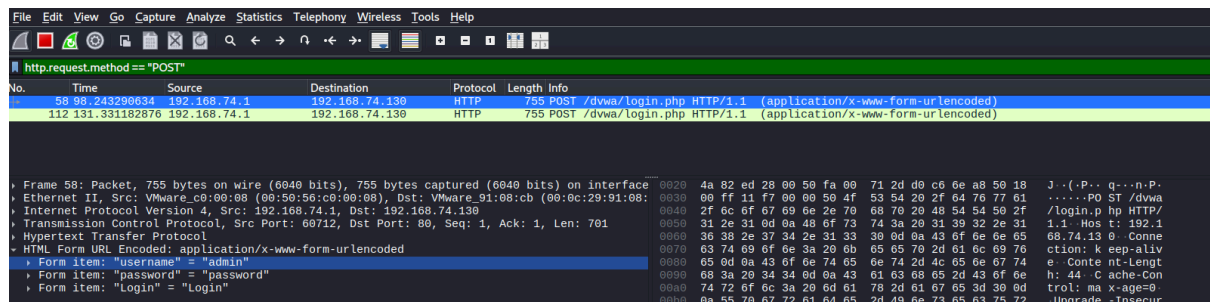
Step 4: Filter HTTP Traffic

Use one of the following display filters:

http

or

http.request.method == "POST"



```
HTML Form URL Encoded: application/x-www-form-urlencoded
  Form item: "username" = "admin"
  Form item: "password" = "password"
  Form item: "Login" = "Login"
```

Step 5: Locate the Login Request

Look for an HTTP **POST** request directed to the DVWA login page (such as [login.php](#)).

Select the packet to inspect its details.

Step 6: Inspect the Packet

Expand the protocol layers in the packet details pane:

- Ethernet
- IP
- TCP
- HTTP

Observe the HTTP request structure, including:

- Request Method
- Destination

- Headers
- Form data (if present)

Step 7: Follow the HTTP Stream

Right-click the HTTP packet and choose:

Follow → HTTP Stream

This displays the complete HTTP conversation, illustrating that the request contents are sent without transport-layer encryption.

Step 8: Save the Capture

Save the packet capture as a **.pcapng** file for documentation and later analysis.

Step 9: Capture Evidence

Take screenshots of:

- Wireshark capturing packets
- HTTP POST request
- Packet details pane
- Follow HTTP Stream window
- Successful DVWA login page

Expected Result

The capture demonstrates that when a web application uses **HTTP instead of HTTPS**, login requests are not protected by TLS. Anyone with authorized access to the traffic (for example, capturing their own traffic in a lab or analyzing a provided packet capture) can inspect the HTTP request contents, illustrating why HTTPS is essential for protecting credentials in transit.